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Effect of limited air exposure and comparative performance between thermophilic and mesophilic solid-state anaerobic digestion of switchgrass

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HIGHLIGHTS

- Mesophilic and thermophilic solid-state anaerobic digestion (SS-AD) of switchgrass.
- Limited air exposure to SS-AD digester headspace did not affect SS-AD performance.
- Increasing TS from 20% to 30% negatively affected volumetric methane productivity.
- Thermophilic SS-AD had about 30% greater methane yield than mesophilic SS-AD.
- Optimal digestion periods for SS-AD of switchgrass are 25–35 days.

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ABSTRACT

Switchgrass is an attractive feedstock for biogas production via anaerobic digestion (AD). Many studies have used switchgrass for liquid anaerobic digestion (L-AD), but few have used switchgrass for solid-state anaerobic digestion (SS-AD). Limited air exposure to the reactor headspace has been adopted in commercial scale anaerobic digesters for different applications. However, little research has examined the effect of limited air exposure on biogas production during SS-AD. In this study, the effects of air exposure and total solids (TS) content on SS-AD performance were evaluated under mesophilic (36 ± 1 °C) and thermophilic (55 ± 0.3 °C) conditions. Limited air exposure did not significantly influence the methane yield during SS-AD. Thermophilic SS-AD had greater methane yields (102–145 L CH_4 kg^{-1} VS_{added}) than mesophilic SS-AD (88–113 L CH_4 kg^{-1} VS_{added}). Both mesophilic SS-AD (73–136 GJ) and thermophilic SS-AD (2–95 GJ) produced positive net energy based on a theoretical 'garage-type' SS-AD digester operating in a temperate climate.

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1. Introduction

The United States Department of Energy (U.S. DOE) considers switchgrass (*Panicum virgatum* L.) as a model lignocellulosic energy crop, due to its high productivity, efficient water and nutrient use, and adaptability to marginal lands (Keshwani and Cheng, 2009). Anaerobic digestion (AD) is a robust process that is able to convert complex organic material such as energy crops into biogas composed of 60–70% methane (Barbanti et al., 2014). Biogas is a flexible renewable fuel that can be upgraded into transportation fuels or used directly to generate electricity and heat (Li et al., 2011). In fact, biogas from AD has recently been included in the United States Renewable Fuels Standard (US EPA, 2014), and switchgrass has been tested as a feedstock in liquid-AD (L-AD) systems (Frigon et al., 2012; Massé et al., 2010), which use low total solids (TS) contents of less than 15% (Xu et al., 2014). However, floating and stratification of fibrous materials has made L-AD of lignocellulosic biomass difficult to scale up (Frigon and Guiot, 2010). In contrast, solid-state anaerobic digestion (SS-AD) systems operate at TS contents greater than 15%. Compared to L-AD, SS-AD has higher volumetric productivities (Li et al., 2011) and generates a digestate that is easier to transport due to low moisture (Xu et al., 2014). These advantages make SS-AD an intriguing option for bioenergy production from lignocellulosic feedstocks, such as switchgrass. However, research on SS-AD of switchgrass is limited (Ahn et al., 2010; Brown et al., 2012; El-Mashad, 2013). Prior to the scale up of this promising approach, it is essential to evaluate the performance of SS-AD of

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switchgrass under different environmental conditions, such as limited air exposure and operating temperature.

A common practice in L-AD systems is to supply oxygen (O_2) or air to the digester headspace during operation. Limited aeration may lead to increased rates of hydrolysis and fermentation, likely because many fermentative AD microbes are facultative anaerobes (Botheju and Bakke, 2011). In fact, several studies showed that limited air addition did not inhibit the strictly anaerobic methanogens during L-AD, and actually enhanced methane yield (Díaz et al., 2010; Lim and Wang, 2013). In addition, limited aeration is also utilized to remove hydrogen sulfide (H_2S) in the biogas prior to upgrading (Díaz et al., 2010). Therefore, limited air exposure could be a helpful process to improve both biogas production and quality in SS-AD. However, there is little research available on the effect of air exposure on biogas production during SS-AD (Charles et al., 2009).

SS-AD can operate under either mesophilic (37 °C) or thermophilic (55 °C) conditions. Mesophilic AD is better established, and often more stable and reliable than thermophilic AD (Labatut et al., 2014). However, thermophilic conditions have been shown to accelerate the conversion of organic material into biogas during SS-AD (Li et al., 2011). Thermophilic temperatures may result in inhibitory levels of volatile fatty acids (VFAs) due to enhanced activity of fermentative microbes (Ahn et al., 2010; Labatut et al., 2014; Shi et al., 2013), which can be addressed by controlling the feedstock to inoculum (F/I) ratio and carbon to nitrogen (C/N) ratio, and providing proper nutrients and pH buffering (Lin et al., 2014; Shi et al., 2014). One major concern about thermophilic SS-AD is the high energy input required to maintain the thermophilic temperature, which may vary significantly for different seasons and can offset the high biogas production rates (Li et al., 2011). To the best knowledge of the authors, there have been no reports that evaluated the performance between mesophilic and thermophilic SS-AD of switchgrass based on energy inputs and outputs while operating at elevated TS contents (>20%).

To address these research gaps, the objectives of this study were to: (1) evaluate the effect of limited air exposure on biogas production during SS-AD; (2) compare performance of mesophilic and thermophilic SS-AD of switchgrass at elevated TS contents; and (3) determine the net energy production during mesophilic and thermophilic SS-AD of switchgrass during long-term operation.

2. Methods

2.1. Feedstock and inoculum

Switchgrass (*Panicum virgatum L.*, cultivar: Cave-in-Rock) was collected in October 2009 from a farm located at the Ohio Agricultural Research and Development Center (OARDC) in Jackson, Ohio. All experiments were done at the OARDC in Wooster, Ohio. Upon receipt, switchgrass was oven dried at 40 °C in a convection oven (Precision Thelco Model 18, Waltham, MA, USA) to less than 10% moisture, ground with a hammer mill (Mackisik, Parker Ford, PA, USA) to pass through a 5 mm screen, and stored in air-tight containers prior to use. Switchgrass was composed of 91.3 ± 0.2% TS, 96.9 ± 0.0% volatile solids (VS, based on TS), 3.1 ± 0.0% ash (based on TS), 43.7 ± 0.7% total carbon (TC), and 0.6 ± 0.1% total nitrogen (TN). The extractives, cellulose, hemicellulose, and lignin contents were 12.1 ± 1.2%, 31.0 ± 1.0%, 19.5 ± 0.6%, and 19.3 ± 0.4%, respectively (all based on TS of sample).

Raw effluent (TS = 7.7%) was collected from a commercial scale, mesophilic liquid anaerobic digester (KB Compost Services, Akron, OH, USA) which uses municipal sewage sludge as a feedstock. Raw effluent was dewatered by centrifugation to increase the TS from 7.7% to 17.8%. Raw and centrifuged effluents were stored in air-tight buckets at 4 °C prior to use. Proportions of raw and centrifuged effluent were activated anaerobically for one week at 36 ± 1 and 55 ± 0.3 °C prior to inoculation in mesophilic and thermophilic SS-AD reactors, respectively.

2.2. Solid-state anaerobic digestion

The effects of TS, temperature, and air exposure on the performance of SS-AD were evaluated using a three factor-two level (2^3) experimental design. Two levels of TS (20%, 30%), temperature (36 ± 1, 55 ± 0.3 °C), and air exposure (no exposure, limited exposure) were used for the SS-AD experiments. Three replicate reactors were designed for each treatment combination (24 total reactors). The raw L-AD effluent, centrifuged L-AD effluent, and switchgrass were thoroughly mixed to the desired TS at a feedstock to inoculum (F/I) ratio of 3 (based on VS). For reactors at 20% TS, switchgrass, raw effluent, and centrifuged effluent represented 64%, 31%, and 8% of the reactor content (dry basis), respectively. For reactors at 30% TS, switchgrass, raw effluent, and centrifuged effluent represented 64%, 8%, and 28% of the reactor content (dry basis), respectively. Pre-mixing was conducted under aerobic conditions. SS-AD was carried out in 1-L glass reactors, and each was loaded with 400 g (30% TS) to 550 g (20% TS) of mixed materials to reach an initial working volume of ~850 mL. After loading, half of the reactors were purged with pure nitrogen (N_2) gas for 1 min to induce oxygen-depleted conditions (initial O_2 content < 0.3%), then sealed with a rubber stopper. The remaining reactors were simply sealed with a rubber stopper after mixed materials were loaded. All reactors were then placed in either a 36 ± 1 °C or 55 ± 0.3 °C incubator for mesophilic and thermophilic SS-AD, respectively. The full digestion period was 70 days. On days 10, 20 and 30, 100 mL of headspace gas was displaced by ambient air using a plastic syringe. This was only conducted for unpurged reactors. Biogas was collected in 5-L Tedlar gas bags (CEL Scientific, Santa Fe Springs, CA) attached to a single outlet on each reactor. For the first 35 days, biogas composition and volume were measured every 2–4 days. From day 35 to day 55, biogas composition and volume were measured once per week. As controls, reactors loaded only with centrifuged L-AD effluent were run in parallel. One of the N_2 purged reactors under thermophilic conditions (TS = 20%) had seal failure on day 15, and was not included in data analysis.

2.3. Analytical methods

Compositional analysis of switchgrass, raw L-AD effluent, centrifuged L-AD effluent, mixed materials before SS-AD, and digestate after SS-AD was conducted. The TS, VS, pH, and alkalinity were measured according to Standard Methods for the Examination of Water and Wastewater (APHA, 2005). Samples for volatile fatty acid (VFA) measurement were prepared using an adapted version of the methods described by Shi et al. (2013), which consisted of suspending 5 g of L-AD effluent or digestate in 5 mL of deionized water, thoroughly mixing it, and separating the solids by centrifugation (10,000 rpm for 5 min). The supernatant was acidified to a pH of 2–3 by addition of hydrochloric acid, and then filtered via a syringe filter (0.2 μm). VFAs (acetic, propionic, isobutyric, butyric, isovaleric, valeric acids) were measured using a gas chromatograph (GC) (Shimadzu, 2010 PLUS, Columbia, MD, USA) equipped with a 30 m × 0.32 mm × 0.5 μm Stabilwax polar phase column and flame ionization detector. Total carbon (TC) and total nitrogen (TN) were measured with an elemental analyzer (Vario Max CNS, Elementar Americas, Mt. Laurel, NJ, USA) in order to calculate the C/N ratio. Total ammonia nitrogen (TAN), composed of free ammonia (NH_3) and ammonium (NH_4^+), was measured by a modified distillation and titration method (ISO 5664, 1984) that used 4% boric acid and a Kjeldahl Distillation Unit B-324 (Buchi, Labortechnik, AG, Switzerland). Extractives of raw materials and SS-AD digestate were measured according to the NREL Analytical Procedure

described in Sluiter et al. (2005). Structural carbohydrates (cellulose, hemicellulose, lignin) were analyzed according to the NREL Analytical Procedure described in Sluiter et al. (2008). Extractives-free biomass was hydrolyzed into monomers after a two-step acid hydrolysis, and the concentrations of these monomers were determined by high performance liquid chromatography (HPLC) (Shimadzu LC-20AB, Columbia, MD, USA) (Sluiter et al., 2008). Acid soluble lignin was measured by UV–Vis spectroscopy, and acid insoluble lignin was determined by gravimetric analysis (Sluiter et al., 2008).

Biogas in Tedlar bags was adapted to ambient pressure and ~25 °C before measurement of composition and volume. The composition of biogas (CO_2, CH_4, N_2, and O_2) was analyzed by a GC (HP 6890, Agilent Technologies, Wilmington, DE, USA) equipped with an alumina/KCl deactivation column (30 m × 0.53 mm × 10 μm) and a thermal conductivity detector. Helium was used as a carrier gas at a flow rate of 5.2 mL min^{-1}. Temperatures of the injector, column, and detector were set at 150, 40, and 200 °C, respectively. The volume of biogas was measured with a drum-type gas meter (Ritter, TG 5, Bochum, Germany). Immediately following the 100 mL air displacement on days 10, 20, and 30, headspace gas was also subjected to compositional analysis by GC. Performance of SS-AD was evaluated using the daily CH_4 production (L kg VS_{added}^{-1} d^{-1}), cumulative CH_4 yield (L kg VS_{added}^{-1}), and volumetric CH_4 productivity ($L_{methane}/L_{work}$). Cumulative CH_4 yield was calculated as the total CH_4 production over the digestion period minus that produced by the inoculum. Volumetric CH_4 productivity was expressed as $V_{methane}/V_{work}$, where $V_{methane}$ was the volume of methane produced and V_{work} was the working volume of the reactor (Brown et al., 2012).

2.4. Energy analysis

Net energy analysis was used to determine whether the useful energy in the biogas could offset the environmental heat losses in a pilot-scale batch SS-AD digester operating outdoors in a temperate U.S. climate in six successive 35 day digestion periods over seven months. The daily net energy (Eq. (1)) was calculated for each reactor configuration (temperature, TS) evaluated in this study.

$$\textit{Daily NetEnergy}\ (j/d) = \textit{Biogas energy} - \textit{Heat required to maintain temperature} \quad (1)$$

Biogas energy was calculated by multiplying the measured daily CH_4 yields by the lower heating value of CH_4 at 25 °C and 101 kPa (33 MJ m^{-3}) and an assumed combined heat and power unit total efficiency of 0.6 (US EPA, 2013). Heat required to maintain the designated temperature was equal to heat losses minus heat generated by microbial activity. Heat generated by microbial activity was determined using Eq. (2) (Lindorfer et al., 2006):

$$\textit{Heat generated by microbial activity}\ (j/d) = M_E * \Delta T_{SH} * 4200 \quad (2)$$

where M_E is the mass of effluent in the digester (kg), ΔT_{SH} is the daily temperature increase in energy crop digesters due to microbial self-heating measured by Lindorfer et al. (2006) (used 0.15 °C d^{-1}), and 4200 is the specific heat of water (J kg^{-1} $°C^{-1}$). Heat losses were calculated using Eq. (3):

$$\textit{Heat losses}\ (j/d) = U * A * \Delta T \quad (3)$$

where U is the coefficient of heat transfer (J m^{-2} d^{-1} $°C^{-1}$); A is the cross sectional area through which heat loss is occurring (m^2), and ΔT is the temperature drop across the surface (°C). Radiative heat transfer was not included. Energy required for initial material heating was included in the cumulative net energy production on the days when the digester contents were removed and new switchgrass and inoculum were introduced to the digester. Energy required for material heating was calculated by multiplying the specific heat of water (4200 J kg^{-1} $°C^{-1}$) and the initial moisture content of the digester contents. It was assumed that feedstock and inoculum are premixed at 20 °C prior to loading into the digester.

The common "garage-type" SS-AD digester was used for this theoretical evaluation. The digester was made of 300-mm thick concrete and had dimensions of 10 m × 5 m × 5 m ($L \times W \times H$). The walls were all insulated and floors were in contact with dry earth (Metcalf & Eddy, 2003). A floating cover with a 25-mm insulating board installed under roofing was used for the ceiling. Heat transfer coefficients for digester construction materials were obtained from Metcalf & Eddy (2003).

The switchgrass was assumed to be harvested in October 2013 and was dried and stored until April 2014. The theoretical scaled up SS-AD digester was assumed to be operated in successive 35 day digestion periods during the months of April–November. After each 35 day digestion period, digester contents were removed, new switchgrass and inoculum were heated to the required temperature (mesophilic (37 °C) or thermophilic (55 °C)) over two days and digested for another 35 days. The selection of 35 day digestion periods was due to observation of rapid decline in CH_4 productivity and net energy production after day 35. Since the switchgrass was collected from Jackson, Ohio, and it was assumed that the theoretical scaled up digester would be operating at the location where the switchgrass was grown, local air and soil temperature data (2014) from the website of the OARDC weather station in Jackson, Ohio (39.0519°, −82.6367°) were used to determine daily heat losses for the seven months of operation.

2.5. Statistical analysis

Statistical significance was determined by analysis of variance (ANOVA, $\alpha = 0.05$) or Tukey's Honestly Significant Difference (HSD) test using JMP Statistical Software from SAS Institute Inc. (Version 10.0.2, Cary, NC, USA). Experimental data were presented as average values ± standard deviations.

3. Results and discussion

3.1. Composition of switchgrass and L-AD effluent

Switchgrass had high TS (91.3%), VS (96.9% of TS), and C/N ratio (79.4). High TS content, high F/I ratio, and high C/N ratio may cause hydrolytic inhibition, rapid accumulation of VFAs, and/or nitrogen limitation during SS-AD (Abbassi-Guendouz et al., 2012; Shi et al., 2014). In fact, Ahn et al. (2010) observed rapid accumulation of VFAs (12 g L^{-1}), low biogas production (2.5 L over 60 days), and low CH_4 content in the biogas (max of 24.5% CH_4) at a switchgrass to inoculum (dairy manure and inoculum) ratio of 4.9 (based on VS), C/N ratio of 39, and TS of 15%. In the present study, switchgrass also had substantial lignin content (19.3%), which has been shown to be recalcitrant to cellulolytic degradation (Meng and Ragauskas, 2014). Despite these compositional limitations, a switchgrass to inoculum (L-AD effluent) ratio (F/I) of 3 provided stable initial conditions for SS-AD. Stability was ensured by the alkalinity (~14 g kg^{-1} as $CaCO_3$) provided by the L-AD effluent, which protected reactors against acidification through pH buffering (Shi et al., 2014). Concurrently, L-AD effluent had a low C/N ratio (Table 1), which provided the SS-AD reactor mixtures with balanced initial C/N ratios from 15 (TS = 20%) to 20 (TS = 30%). L-AD effluent activated at 55 ± 0.3 °C had significantly higher VFAs because higher temperatures enhance the growth kinetics of acidogenic microbes in AD (Labatut et al., 2014). This caused initial VFAs in thermophilic reactors (2.6–3.8 g kg^{-1}) to be 2–3 times

higher than the initial VFAs for mesophilic reactors (1.3–1.4 g kg^{-1}) (Table 2).

3.2. Effect of air exposure on SS-AD performance

Immediately following air exposure on days 10, 20 and 30, there was elevated N_2 content (19–28%) and O_2 content (5–8%) in the reactor headspace. This briefly diluted the CH_4, which contributed to lower daily CH_4 production on the days following the air exposure (Fig. 1c). However, methanogenic O_2 inhibition was not observed, as CH_4 content quickly recovered (Fig. 1c) to levels similar to reactors pre-purged with N_2 (Fig. 1a). Overall, 70-day cumulative CH_4 and CO_2 yields were not significantly affected by air exposure (ANOVA, $p > 0.05$). This was probably due to formation of diffusion barriers between the "sticky" contents of SS-AD and the gas phase with O_2 introduced via air displacement (Botheju and Bakke, 2011). It is most likely that air displacement simply diluted the CH_4 in the biogas and the O_2 introduced could not diffuse into reactor materials to cause inhibition even though the measured content of O_2 after air addition in this study (2.2–3.6 mM) was above the inhibitory oxygen content (30 nM) for methanogens in anaerobic digesters reported by Scott et al. (1983). These results are significant because SS-AD performance may not be impacted by low air exposure levels. Further studies could be conducted to evaluate SS-AD performance at higher air exposure levels. Presence of H_2S content in the biogas was not detected, likely due to the limited sulfur content of switchgrass. Because reactors purged with N_2 and reactors with limited air exposure had similar CH_4 (Fig. 1a and c) and CO_2 yields (Fig. 1b and d), reactors with limited air exposure were not included in the net energy analysis and analysis of organic component degradation.

3.3. Effects of temperature and TS on SS-AD performance

Thermophilic conditions improved SS-AD performance. At 20% TS, thermophilic reactors had daily biogas production peaks on day 2 (12.7 L kg^{-1} VS d^{-1}) and day 12 (17.6 L kg^{-1} VS d^{-1}), while mesophilic reactors had 23% and 48% lower peaks on day 2 and day 8, respectively. The first biogas peak on day 2 was primarily composed of CO_2, and peak daily CO_2 production was nearly two times greater under thermophilic conditions (Fig. 1b). The second biogas peaks were due primarily to CH_4 production, and thermophilic reactors had nearly three times higher peak daily CH_4 production than mesophilic reactors (Fig. 1a). Initial peak CO_2 production was likely due to enhanced hydrolysis and acidogenesis, which also caused VFA accumulation and imbalance between acidogenesis and methanogenesis (Shi et al., 2013). This imbalance also likely contributed to a longer lag phase for cumulative CH_4 production (Fig. 2). Additionally, the imbalance could also have been heightened by higher initial VFAs in thermophilic reactors (Table 2). However, alkalinity increased and pH remained fairly stable before and after the digestion period (Table 2), which indicated that buffering was sufficient. Furthermore, methanogenic activity was maintained under all operating conditions, as indicated by the CH_4 content in the biogas, which was stable (55–60%) throughout the digestion period (Fig. 1a).

Enhanced hydrolysis was observed for reactors under thermophilic conditions. In fact, at both TS contents, reactors under thermophilic conditions had significantly higher ($p < 0.05$) holocellulose (cellulose + hemicellulose) degradation than reactors under mesophilic conditions (Table 3). Aside from a more active population of cellulolytic and xylanolytic microbes (Shi et al., 2013) and enhanced reaction kinetics (Labatut et al., 2014), increased temperatures could have improved the mass transfer of hydrolysis products to AD microflora (Ribeiro et al., 2006). The decline in performance under mesophilic conditions and 30% TS was very likely due to hydrolytic inhibition caused by mass diffusion limitations (Xu et al., 2014). Hydrolytic products (sugars) may have accumulated at the substrate surface due to limited mass transfer, eventually causing inhibition to hydrolytic enzyme adsorption (Xu et al., 2014). Mass diffusion limitations could have limited the amount of hydrolytic products available to fermentative microbes, reducing the fermentative products (H_2, acetic acid, CO_2) available for methanogenic conversion. This claim is supported because there was reduced CH_4 production but limited evidence of acidification or ammonia inhibition (Table 2), indicating hydrolysis products were unavailable to fermentative microbes (Abbassi-Guendouz et al., 2012).

There was a significant positive correlation between holocellulose degradation and 35-day cumulative CH_4 yield ($p < 0.05$, $R^2 = 0.76$), which indicates that the degradation of cellulose and hemicellulose in switchgrass positively contributed to CH_4 production. However, there was similar holocellulose degradation in mesophilic reactors at 20% TS and 30% TS, despite higher CH_4 production and VS degradation at 20% TS. Furthermore, there was better correlation between VS reduction and 35-day cumulative CH_4 yield ($p < 0.05$, $R^2 = 0.90$). These discrepancies indicate that other organic components, such as proteins and extractives, also contributed to the total CH_4 production. Before digestion, only ~3% of cellulose and ~7% of hemicellulose in the reactor contents came from the inocula, indicating most of the degraded cellulose and hemicellulose should have come from switchgrass.

Table 1
Composition of L-AD effluent.

Component	Raw effluent (mesophilic)	Centrifuged effluent (mesophilic)	Raw effluent (thermophilic)	Centrifuged effluent (thermophilic)
TS (%)	7.7 ± 0.0[b]	17.9 ± 0.5[c]	7.7 ± 0.0[b]	17.8 ± 0.4[c]
VS (%)[a]	58.6 ± 0.1[b]	57.0 ± 1.0[c]	58.6 ± 0.1[b]	54.3 ± 0.3[c]
Ash content (%)[a]	41.4 ± 0.1[b]	43.1 ± 1.0[c]	41.4 ± 0.1[b]	45.7 ± 0.3[c]
TC (%)	2.9 ± 0.0[c]	6.3 ± 0.2[c]	3.2 ± 0.1[c]	6.4 ± 0.1[c]
TN (%)	0.6 ± 0.1[c]	0.8 ± 0.1[c]	0.6 ± 0.0[c]	0.7 ± 0.0[c]
C/N ratio	5.1 ± 0.5[c]	7.9 ± 0.3[c]	5.7 ± 0.4[c]	8.7 ± 0.1[c]
pH	8.5 ± 0.0[c]	8.7 ± 0.0[c]	8.4 ± 0.0[c]	8.6 ± 0.0[c]
Alkalinity (g kg^{-1} as $CaCO_3$)	14.0 ± 0.8[c]	14.3 ± 0.5[c]	14.9 ± 0.7[c]	13.1 ± 0.2[c]
VFAs (g kg^{-1})	1.0 ± 0.0[c]	1.2 ± 0.0[c]	3.5 ± 0.1 [c]	6.1 ± 0.3[c]
TAN (g kg^{-1})	4.0 ± 0.1[c]	4.6 ± 0.3[c]	4.4 ± 0.2[c]	5.2 ± 0.2[c]
Extractives (%)[a]	14.0 ± 0.2[b]	14.0 ± 1.1[b]	14.0 ± 0.2[b]	14.0 ± 1.1[b]
Cellulose (%)[a]	1.5 ± 0.2[b]	1.5 ± 0.1[b]	1.5 ± 0.2[b]	1.5 ± 0.1[b]
Hemicellulose (%)[a]	2.8 ± 0.6[b]	2.4 ± 0.3[b]	2.8 ± 0.6[b]	2.4 ± 0.3[b]
Lignin (%)[a]	NA	NA	NA	NA

Values presented as average ± standard deviation.
[a] Based on TS of sample; other components based on total weight of sample; NA = not analyzed.
[b] Characteristics of effluent prior to activation.
[c] Characteristics of effluent after 7 days of activation at 36 ± 1 °C (mesophilic) or 55 ± 0.3 °C (thermophilic).

Table 2
Reactor characteristics before and after 70 days of SS-AD.

Temperature (°C)	TS (%)	pH		Alkalinity (g kg^{-1} as $CaCO_3$)		Total ammonia nitrogen (g kg^{-1})		Total VFAs (g kg^{-1})	
		Initial	Final	Initial	Final	Initial[a]	Final	Initial	Final
Mesophilic (36 ± 1 °C)	20	8.3 ± 0.0	8.0 ± 0.1	11.6 ± 0.1	13.9 ± 0.3	3.4	4.4 ± 0.0	1.3 ± 0.1	0.1 ± 0.0
	30	8.0 ± 0.1	8.5 ± 0.1	11.3 ± 0.6	13.1 ± 0.5	3.4	3.9 ± 0.0	1.4 ± 0.4	0.0 ± 0.0
Thermophilic (55 ± 0.3 °C)	20	8.4 ± 0.1	8.2 ± 0.0	11.5 ± 0.1	12.5 ± 0.0	3.8	4.3 ± 0.0	2.6 ± 0.1	1.7 ± 0.3
	30	8.2 ± 0.3	8.4 ± 0.1	10.8 ± 0.8	12.5 ± 0.6	3.9	4.2 ± 0.0	3.8 ± 0.5	3.2 ± 0.7

[a] Calculated from feedstock composition.

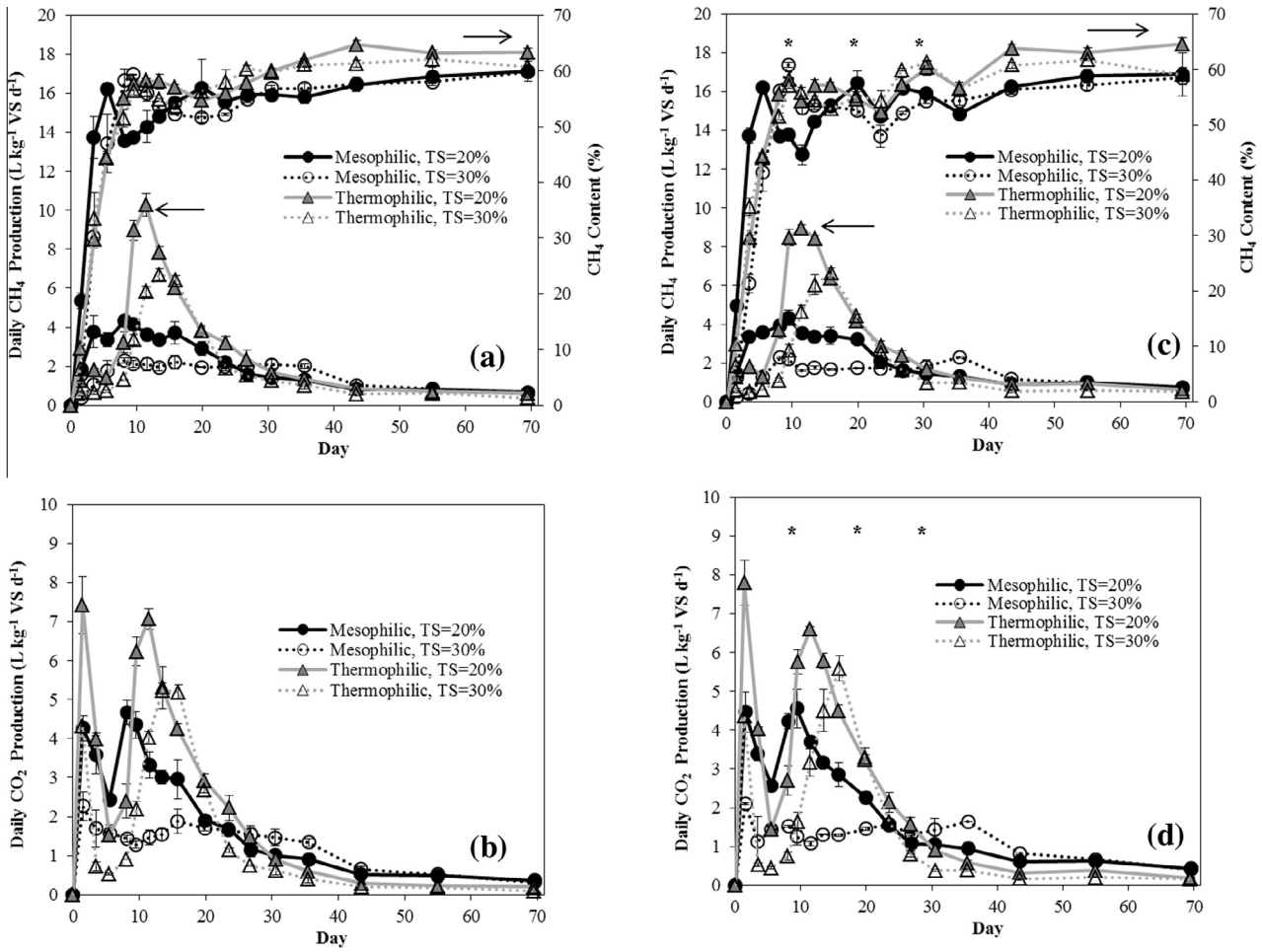


Fig. 1. Effects of reactor conditions on SS-AD performance for N_2 purged reactors (a, b) and reactors with limited air exposure (c, d). Asterisks show days when 100 mL of air was displaced into the headspace.

3.4. Performance perspectives and net energy analysis

Thermophilic conditions were most favorable for the total production of CH_4 because at equivalent TS contents, thermophilic reactors had ~20–30% higher cumulative CH_4 yields than mesophilic reactors (Fig. 2). The 70-day cumulative CH_4 yields for thermophilic SS-AD with 20% TS (145 L CH_4 kg^{-1} VS) in the current study were similar to the 40-day yields reported at 16% TS, F/I ratio of 2, and 50 °C (167 L CH_4 kg^{-1} VS) (El-Mashad, 2013). The 70-day cumulative CH_4 yields from mesophilic reactors at 20% TS (113 L CH_4 kg^{-1} VS) were 41–63% lower than those reported in batch mesophilic L-AD of ensiled switchgrass (191–309 L CH_4 kg^{-1} VS) (Massé et al., 2010) but similar to mesophilic SS-AD of switchgrass at 18% TS and F/I ratio of 3 (117 L CH_4 kg^{-1} VS) (Brown et al., 2012). The lower CH_4 yields of this study compared to Massé et al. (2010) can be ascribed to higher TS, a higher F/I ratio, and no pretreatment of switchgrass prior to SS-AD. CH_4 production was very slow after day 35. In fact, about 70–78% (mesophilic) to 85–86% (thermophilic) of the cumulative CH_4 yield at 70 days was produced by day 35 (Fig. 2). These results indicate that a shorter digestion period could be more advantageous for the SS-AD conditions tested.

One main advantage of operating SS-AD at increased TS content is the ability to provide more feedstock per unit volume. However, an increased proportion of low bulk density switchgrass reduced the bulk density of the reactor content from 0.70 to 0.45 kg L^{-1} when TS content was increased from 20% to 30%. Thus, volumetric methane productivity was lower for reactors at 30% TS (Fig. 3). Still, there was significantly higher ($p < 0.05$) volumetric CH_4 production

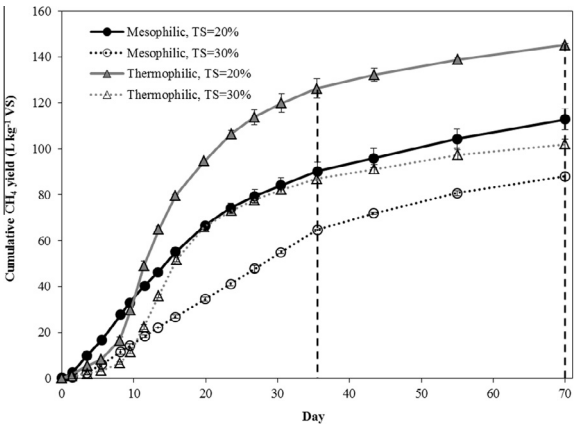


Fig. 2. Cumulative CH_4 yield for N_2 purged reactors.

Table 3
Cellulose, hemicellulose and VS removal after 70 days of SS-AD.

Temperature (°C)	TS (%)	Cellulose removal (%)	Hemicellulose removal (%)	VS removal (%)
Mesophilic (36 ± 1)	20	32.5 ± 1.9	34.8 ± 2.6	25.2 ± 2.2
	30	30.4 ± 3.6	33.3 ± 4.2	21.5 ± 1.0
Thermophilic (55 ± 0.3)	20	52.7 ± 6.8	58.4 ± 2.7	30.2 ± 3.2
	30	37.6 ± 1.7	43.7 ± 2.4	25.2 ± 0.5

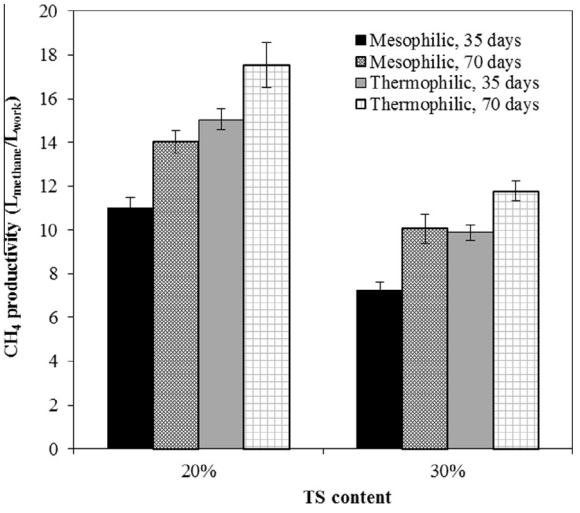


Fig. 3. Comparison of CH_4 productivity for N_2 purged reactors at 35 and 70 days.

from thermophilic reactors than mesophilic reactors. In fact, there was no significant difference ($p > 0.05$) between 35-day CH_4 productivity from thermophilic reactors and 70-day CH_4 productivity from mesophilic reactors. Therefore, an equivalent amount of CH_4 could be produced in thermophilic SS-AD in a shorter digestion period than mesophilic SS-AD. Besides, thermophilic digestion provides greater destruction of pathogens, improving safety of downstream digestate utilization (Labatut et al., 2014).

Table 4 shows the assumptions used in net energy analysis. At equivalent F/I ratios, reactor contents at 30% TS had ~50% higher VS contents than reactors at 20% TS. However, the low bulk density of switchgrass limited the advantage of operating the "garage type" SS-AD digester at 30% TS, because less total material weight could be loaded. Therefore, reactors at 20% TS and 30% TS had nearly equivalent initial VS loading (~23,000 kg VS). In this analysis, the advantage for operation at 30% TS was reduced initial heating of reactor contents, due to lower moisture content. In fact, under both operating temperatures, reactors at 30% TS had 44% lower initial heating requirements.

Table 4
Assumptions used in net energy analysis.

Assumption	Value	Unit
Digestion time		
Digestion period	35	days period^{-1}
Number of digestion periods	6	periods
Total time of digester operation	220	days
Digester design		
Length	10	m
Width	5	m
Height	5	m
Working volume	80	% of total volume
Wall[a] thickness	300[d]	mm
Floor[b] thickness	300[d]	mm
Floating cover[c]	25[d]	mm
U_{wall}	0.8[d]	J m^{-2} s^{-1} °C^{-1}
U_{floor}	1.7[d]	J m^{-2} s^{-1} °C^{-1}
U_{cover}	1.0[d]	J m^{-2} s^{-1} °C^{-1}
Input material properties		
Bulk density (TS = 20%)	0.70[e]	kg FW L^{-1}
Bulk density (TS = 30%)	0.45[e]	kg FW L^{-1}
Moisture content (TS = 20%)	800[e]	g kg^{-1} FW
Moisture content (TS = 30%)	700[e]	g kg^{-1} FW
Temperature	20	°C
Climate conditions (averages)		
Outside air temperature, Apr–May	13.0[f]	°C
Soil temperature, 2", Apr–May	11.9[f]	°C
Outside air temperature, May–Jun	17.3[f]	°C
Soil temperature, 2", May–Jun	17.6[f]	°C
Outside air temperature, Jun–Jul	22.2[f]	°C
Soil temperature, 2", Jun–Jul	24.1[f]	°C
Outside air temperature, Jul–Aug	21.7[f]	°C
Soil temperature, 2", Jul–Aug	23.7[f]	°C
Outside air temperature, Aug–Sept	22.2[f]	°C
Soil temperature, 2", Aug–Sept	23.6[f]	°C
Outside air temperature, Sept–Oct	18.5[f]	°C
Soil temperature, 2", Sept–Oct	20.8[f]	°C
Outside air temperature, Oct–Nov	15.3[f]	°C
Soil temperature, 2", Oct–Nov	14.9[f]	°C

[a] Concrete with insulation.
[b] Concrete in contact with dry earth.
[c] Insulating board installed on roof under floating cover.
[d] Value obtained from Metcalf & Eddy (2003).
[e] Measured value in this study.
[f] OARDC Weather Station, Jackson, OH FW = fresh weight; DW = dry weight.

Net energy analysis showed that all reactor conditions produced positive cumulative net energy. Although initial heating of reactor contents were reduced, there was little incentive to operate at higher TS, because of lower CH_4 production. In fact, thermophilic operation at 30% TS consumed more energy than what was produced by SS-AD in the first 35 day digestion period (Fig. 4b). Daily net energy production was fairly stable throughout the seven month period (Fig. 4a), due to similar ambient temperatures (Table 4).

Operation under thermophilic conditions produced over two times higher peak daily net energy than when operated under mesophilic conditions (Fig. 4a). However, mesophilic conditions at 20% TS had 43% higher peak cumulative net energy production than thermophilic conditions at 20% TS because of reduced initial heating requirements after inoculation (Fig. 4b). These results indicate mesophilic SS-AD is more suitable for bioenergy production in a temperate climate. However, there are limitations to the net

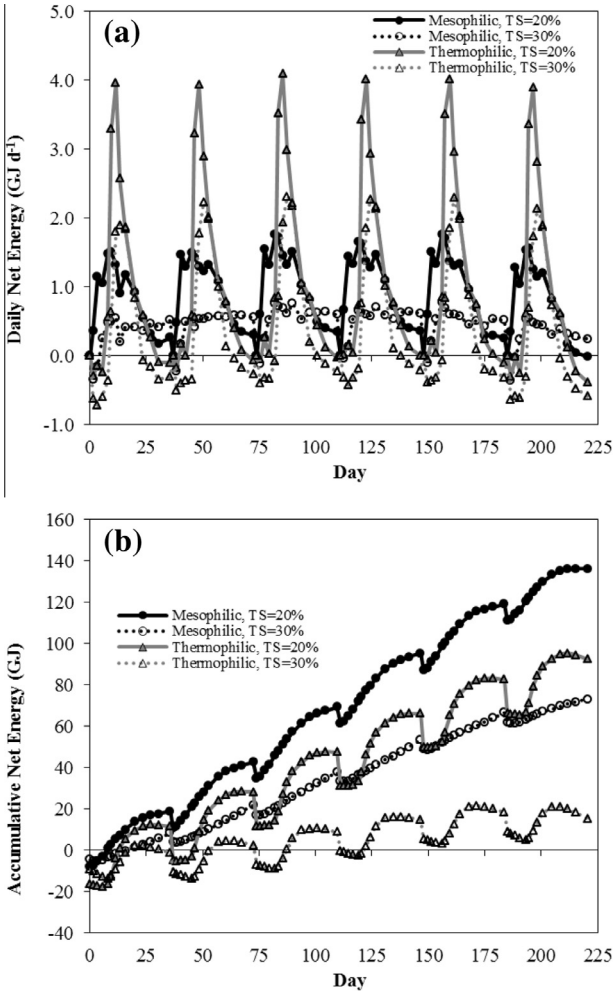


Fig. 4. Daily (a) and cumulative (b) net energy production. *Note:* initial energy for heating is included in (b).

energy analysis that should be addressed in future studies. These include the effect of microbial energy production under thermophilic conditions, optimization of other reactor parameters (F/I ratio, C/N ratio), and effects of harvest time on switchgrass digestibility (Mitchell and Schmer, 2012). Spring harvesting of switchgrass has the advantage of nutrient translocation into the soil and reduced fertilizer costs. However, lower biomass yields are expected during spring (Mitchell and Schmer, 2012). Therefore, integration of energy required for switchgrass cultivation and transportation would improve the energy analysis for SS-AD systems. Design of SS-AD reactors with low surface area to volume ratios and better insulation could reduce heat losses to the environment. Furthermore, chemical or biological pretreatment of switchgrass could also improve biogas production in SS-AD.

4. Conclusions

Minimal air exposure to the digester headspace did not significantly affect reactor performance during the SS-AD of switchgrass. CH_4 yields and organic component degradation were improved during thermophilic SS-AD compared to mesophilic SS-AD, with the highest yields obtained at lower total solids content. The recommended digestion period for SS-AD of switchgrass is 25–35 days, based on the CH_4 productivity and net energy production. SS-AD of switchgrass under mesophilic SS-AD and thermophilic SS-AD produced positive cumulative net energy.

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